

Final Audit Report

2026-05-16

Contents

Final Audit Report	1
Status Distribution	1
Plan Adherence	1
Confidence Calibration	4
Residual Debt	5
Findings by Severity	5
Future Work	6
Reconciliation Log	6
Appendix: Figure Coverage	6

Final Audit Report

Run id: run-2026-05-15T153635Z Document stage timestamp: 2026-05-16T15:38:04Z

Status Distribution

Counts below use the latest event for every milestone-like ledger id in `promise_ledger.jsonl`, including plan, archive, and run-control records. Plan adherence below is restricted to M* milestones from `plan_of_record.md`.

Unified status	Latest ledger milestone count
validated	76
in-progress	1

Validator status at document stage: `promise_check rc=0`, `org_check rc=0`; combined status green.

Plan Adherence

Plan milestones parsed: 39. Missing from ledger: 0. Non-terminal plan milestones at end: 0.

Milestone	Terminal/latest status	Confidence	Existing evidence refs	Latest artifact refs
M1-paper-map	validated	provisional	4	4
M2-proof-ledger	validated	provisional	9	9
M3-computational-probes	validated	provisional	4	4
M4-formal-certification	validated	provisional	6	6

Milestone	Terminal/latest status	Confidence	Existing evidence refs	Latest artifact refs
M5-extension-candidates	validated	provisional	5	5
M6-final-synthesis	validated	provisional	4	5
M7-product-ratio-bounds	validated	provisional	5	5
M8-quotient-family-bridge	validated	provisional	5	5
M9-aggregate-product-ratio-obstruction	validated	provisional	6	6
M10-restricted-quotient-aggregate	validated	provisional	7	7
M11-trace-like-weighted-quotient-class	validated	provisional	9	9
M12-restricted-aggregate-theorem-template	validated	provisional	7	7
M13-cancellation-mechanism-diagnostics	validated	provisional	9	9
M14-external-decay-thresholds	validated	provisional	9	9
M15-kim-tao-bridge-requirement	validated	provisional	7	7
M16-local-spectral-window-corollaries	validated	provisional	8	8
M17-local-window-variance-input	validated	provisional	9	9
M18-test-function-localization-feasibility	validated	provisional	8	8
M19-smoothed-window-paley-wiener-lemma	validated	provisional	10	10
M20-long-support-trace-variance-requirement	validated	provisional	8	8

Milestone	Terminal/latest status	Confidence	Existing evidence refs	Latest artifact refs
M21-trace-side-long-support-variance-template	validated	provisional	8	8
M22-trace-corollary34-uniform-coefficient-variation-target	validated	provisional	8	8
M23-localized-trace-numerator-quotient-family-model	validated	provisional	9	9
M24-localized-transform-geodesic-weight-decay-obstruction	validated	provisional	8	8
M25-local-window-route-synthesis-and-branch-decision	validated	provisional	9	9
M26-post-local-extension-reprioritization	validated	provisional	9	9
M27-multiplicity-and-cluster-corollaries-from-rigidity	validated	provisional	6	6
M28-theorem2-lp-mass-distribution-corollaries	validated	provisional	7	7
M29-pretrace-local-mass-intermediate-from-theorem2-proof	validated	provisional	9	9
M30-schreier-benchmark-theoremization	validated	provisional	9	9
M31-schreier-variance-mechanism-theoremization	validated	provisional	11	11

Milestone	Terminal/latest status	Confidence	Existing evidence refs	Latest artifact refs
M32-schreier-fixed-pair-covariance-lemma	validated	provisional	9	9
M33-schreier-benchmark-package-synthesis	validated	provisional	12	12
M34-finite-nonshrinking-spectral-statistics	validated	provisional	10	10
M35-surface-corollary34-numerator-obstruction	validated	provisional	11	11
M36-direct-small-x-surface-numerator-target	validated	provisional	11	11
M37-signed-pointwise-cancellation-surface-aggregate	validated	provisional	11	11
M38-surface-native-grouping-problem	validated	provisional	11	11
M39-surface-relation-kernel-spc-probe	validated	provisional	11	11

Confidence Calibration

Validated-event confidence counts are computed over all validated ledger events; latest-confidence counts are computed over latest events per ledger milestone.

Confidence	Validated events	Latest milestones
provisional	100	77

Low-confidence terminal plan states: M1-paper-map, M2-proof-ledger, M3-computational-probes, M4-formal-certification, M5-extension-candidates, M6-final-synthesis, M7-product-ratio-bounds, M8-quotient-family-bridge, M9-aggregate-product-ratio-obstruction, M10-restricted-quotient-aggregate, M11-trace-like-weighted-quotient-class, M12-restricted-aggregate-theorem-template, M13-cancellation-mechanism-diagnostics, M14-external-decay-thresholds, M15-kim-tao-bridge-requirement, M16-local-spectral-window-corollaries,

M17-local-window-variance-input, M18-test-function-localization-feasibility, M19-smoothed-window-paley-wiener-lemma, M20-long-support-trace-variance-requirement, M21-trace-side-long-support-variance-template, M22-trace-corollary34-uniform-coefficient-variation-target, M23-localized-trace-numerator-quotient-family-model, M24-localized-transform-geodesic-weight-decay-obstruction, M25-local-window-route-synthesis-and-branch-decision, M26-post-local-extension-reprioritization, M27-multiplicity-and-cluster-corollaries-from-rigidity, M28-theorem2-lp-mass-distribution-corollaries, M29-pretrace-local-mass-intermediate-from-theorem2-proof, M30-schreier-benchmark-theoremization, M31-schreier-variance-mechanism-theoremization, M32-schreier-fixed-pair-covariance-lemma, M33-schreier-benchmark-package-synthesis, M34-finite-nonshrinking-spectral-statistics, M35-surface-corollary34-numerator-obstruction, M36-direct-small-x-surface-numerator-target, M37-signed-pointwise-cancellation-surface-aggregate, M38-surface-native-grouping-problem, M39-surface-relation-kernel-spc-probe

Residual Debt

Severity	Milestone	Kind	Narrative
MODERATE	M6-final-synthesis	latest_event_missing_artifact_reference	M6-final-synthesis latest ledger event references missing artifact path(s): ['reports/final/final_report.md']. Existing artifacts still provide partial support.
CRITICAL	M36-direct-small-x-surface-numerator-target	success_criteria_artifact_missing	Plan success criteria references missing artifact $p(1/n)/Q_{id}(1/n)$ for terminal milestone M36-direct-small-x-surface-numerator-target.

Findings by Severity

CRITICAL

- M36-direct-small-x-surface-numerator-target / success_criteria_artifact_missing: Plan success criteria references missing artifact $p(1/n)/Q_{id}(1/n)$ for terminal milestone M36-direct-small-x-surface-numerator-target.

MODERATE

- M6-final-synthesis / latest_event_missing_artifact_reference: M6-final-synthesis latest ledger event references missing artifact path(s): ['reports/final/final_report.md']. Existing artifacts still provide partial support.

MINOR

None.

Future Work

- M6-final-synthesis: Restore or regenerate a final report artifact matching reports/final/final_report.* and update the final artifact index so the M6 plan success criteria are directly satisfied.
- M6-final-synthesis: Correct the latest M6 ledger artifact references to point at existing final package artifacts, then rerun promise_check and artifact-reference checks to close the evidence-hygiene gap.

Reconciliation Log

No _plan/, _archive/, or correction reconciliation events are emitted by this final auditor. The retained M6 findings describe unresolved artifact debt rather than a ledger correction the auditor can safely commit.

Appendix: Figure Coverage

Metric	Value
Figure files on disk	104
Figure paths referenced by ledger artifacts	105
Milestones with figure refs	38
Warranted milestones without figures	0
Missing referenced figures	0
Likely orphan figures	0